

Stage 2 Mini-Project: Joshimath Landslide Susceptibility Mapping

One-slide summary for oral interview

Main finding

Adding InSAR LOS velocity slightly improved the Random Forest landslide susceptibility model. The improvement is not very large, but it is useful because the InSAR layer adds information about active ground movement.

Model comparison

Run	AUROC	F1
Without InSAR	0.872	0.828
With InSAR	0.893	0.857

Short interpretation

- High values mainly follow steep valley-side terrain.
- InSAR helps because it shows slow ground movement, not only terrain shape.
- The result should be treated carefully because the InSAR-valid sample size was small.

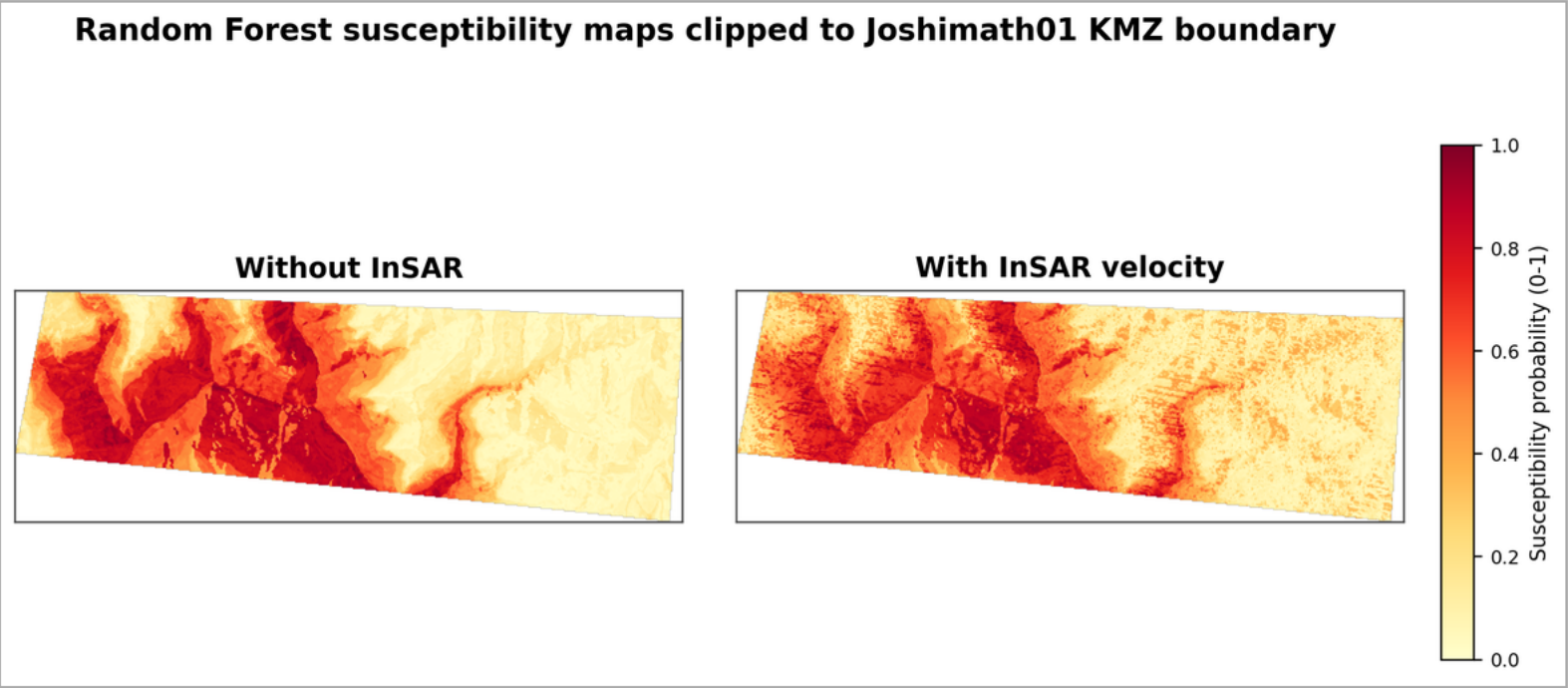


Figure: Susceptibility maps clipped to the Joshimath01 boundary and shown with the same 0-1 scale.